



F1-27 Centrifugal Pump Characteristics

This accessory offers enhanced capabilities provided by the inclusion of a variable-speed pump with inverter drive rather than a fixed-speed pump.

PRODUCT CODE: F1-27

Categories: Educational, F1 Series Fluid Mechanics, F1-27 - Centrifugal Pump Characteristics

Tags: Auxiliary Pump , Centrifugal Pump Characteristics , Diffuser , Discharge Pipe , F1-27 , Flow Control Valve , Fluid Mechanics , Hydraulics Bench , Independent Discharge Manifold , Inverter Drive , Pressure Gauge , Primary Pump , Pump Inlet , Pump Outlet , Variable Speed Pump

DESCRIPTION

This accessory comprises a variable speed pump assembly and independent discharge manifold interconnected by flexible tubing with quick release connectors. This auxiliary pump is intended to be used in conjunction with the basic Hydraulics Bench F1-10/F1-10-2.

The pump speed is varied by an inverter drive. The motor speed, output voltage and motor current can be monitored on the inverter display. A compound pressure gauge is mounted on the pump inlet and a pressure gauge is mounted on the pump outlet.

An independent discharge manifold incorporates a pressure gauge and flow control valve prior to a discharge pipe with diffuser.

The auxiliary pump is mounted on a support plinth designed to be positioned on the floor besides the hydraulics bench , adjustable feet allowing levelling.

TECHNICAL SPECIFICATIONS

Pump: Centrifugal type

Max head : 21m H₂O

Max flow: 1.35 L/s

Motor rating: 0.36kW

Speed controller: Frequency inverter

Speed range: 0-1500 rpm

Pressure gauge range: 0-60m H₂O

Compound gauge range: -10 to +32m H₂O

See Hydraulics Bench F1-10 technical details for primary pump characteristics

FEATURES & BENEFITS

Experimental content

- Determining the relationship between head, discharge, speed, power and efficiency for a centrifugal pump at various speeds
- Determining the head/flow rate characteristics of two similar pumps operating in either parallel or series configuration at the same speed

Requirements

Electrical supply:

- F1-27-A: 220-240V / 1ph / 50Hz @ 10 amp
- F1-27-G: 220V / 1ph / 60Hz @ 10 amp

G version has optional 1.5kVA transformer available to accommodate 120V / 1ph / 60Hz

Requires Hydraulics Bench Service unit F1-10/F1-10-2

Dimensions

Length: 0.36m

Width: 0.16m

Height: 0.325m

Order Codes

- **F1-27-A:** 220-240V / 1ph / 50Hz @ 10 amp
- **F1-27-G:** 220V / 1ph / 60Hz @ 10 amp

YOU MAY ALSO LIKE...



**FM51 – SERIES
AND PARALLEL
PUMPS
DEMONSTRATI
ON**

[Read more](#)



**F1-10
HYDRAULICS
BENCH**

[Read more](#)



**C11-MKII PIPE
NETWORKS**

[Read more](#)



**F1-35 SERIES &
PARALLEL
PUMPS**

[Read more](#)



**C9-MKII
FLOWMETER
DEMONSTRATO
R**

[Read more](#)

Armfield Limited. Unit 10 Headlands Business Park, Salisbury Road, Ringwood, Hampshire BH24 3PB.
England. Tel: +44(0) 1425 478781